

Motherboard Components Explained

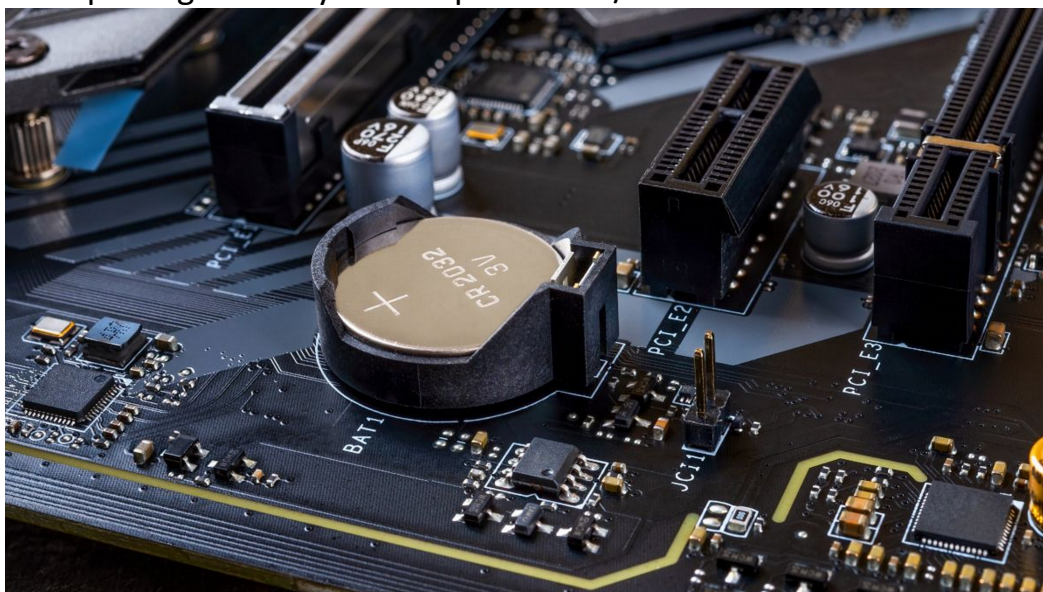
1. CMOS Battery (Complementary Metal-Oxide Semiconductor)

Function:

- Powers a small memory chip that stores BIOS settings (boot order, date/time, hardware configuration).
- Retains BIOS settings even when the PC is turned off or unplugged.

Details:

- Usually a **CR2032 coin cell battery**.
- If removed or dead, system may reset BIOS settings and show wrong time/date.
- Replacing it is easy and helps fix BIOS/boot errors.



2. PCIe x Slots (Peripheral Component Interconnect Express)

Function:

- High-speed interface used to connect expansion cards like **graphics cards, Wi-Fi cards, SSDs**, and more.

Slot Types:

Slot Name Lanes Common Use

PCIe x1	1	Network cards, USB expanders
PCIe x4	4	NVMe SSDs, capture cards
PCIe x8	8	RAID cards, sometimes GPUs
PCIe x16	16	Graphics cards (GPUs)

Version Speeds:

- PCIe 3.0, 4.0, 5.0 – each newer version has double the speed of the previous.



⚡ 3. VRM (Voltage Regulator Module)

Function:

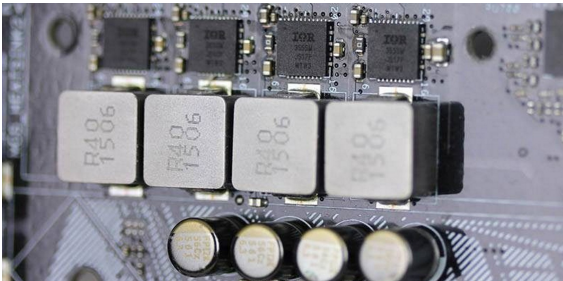
- Converts **12V power** from the SMPS to a stable lower voltage (e.g., 1.2V–1.4V) that the **CPU and RAM require**.
- Maintains power stability and prevents component damage.

Importance:

- A good VRM ensures **stable overclocking** and performance.
- Poor VRMs can cause **CPU throttling or shutdowns** under load.

Components in VRM:

- MOSFETs, chokes, capacitors, and phase controllers.



1.2 V > Processor > CPU 12 V

🔧 4. Chipset

Function:

- Acts as the **communication hub** between the CPU, RAM, GPU, storage, and other peripherals.
- Determines what features your motherboard supports: USB ports, overclocking, NVMe drives, etc.

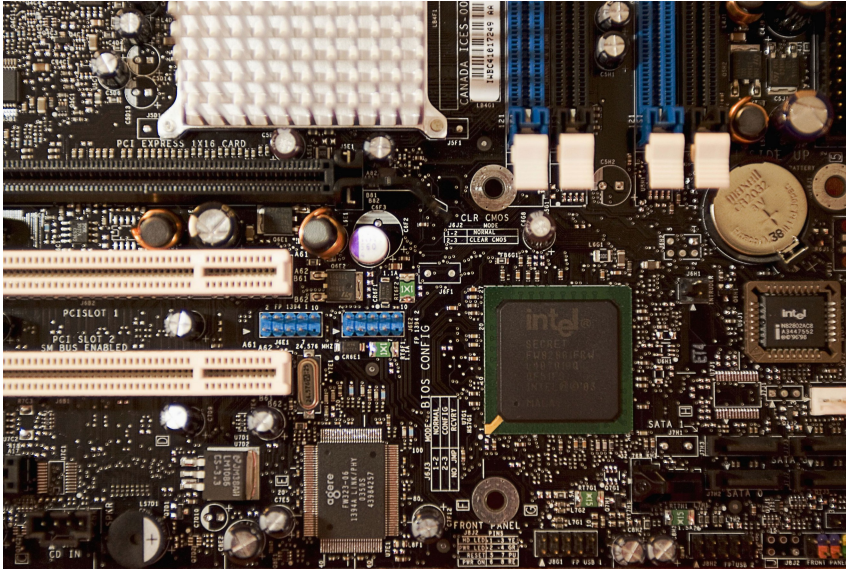
Types:

- **Intel Chipsets:** B760, Z790 (Z supports overclocking).
- **AMD Chipsets:** B550, X670 (X has more features and lanes).

Divided into:

- **Northbridge (older boards):** Handles CPU, RAM, GPU communication.

- **Southbridge:** Manages input/output functions (USB, SATA, etc.)
Modern chipsets are unified, and the CPU handles much of what the northbridge once did.



🔌 5. System Panel Pins (Front Panel Header)

Function:

- Tiny pins on the motherboard that connect to your case's **power button, reset button, LED lights, and speaker.**

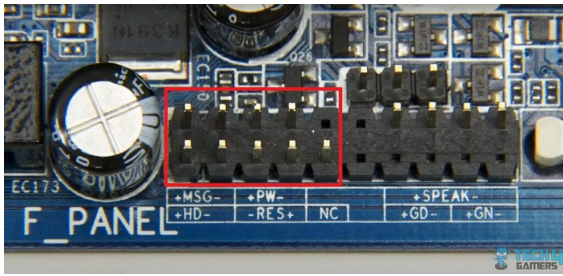
Labelled as:

- **PWR_SW:** Power switch
- **RESET_SW:** Reset switch
- **HDD_LED:** Hard drive activity light
- **PWR_LED+:** Power light
- **SPEAKER:** Beep codes for diagnostics

Installation Tip:

Pins are small—use the **motherboard manual** to connect them correctly.





🔮 6. Crystal Oscillator (Quartz Crystal)

Function:

- Provides a **precise and constant timing signal** (clock signal) to synchronize all components of the motherboard.

Where it's used:

- CPU clocking
- RAM timing
- USB timing
- Real-time clock (RTC)

How it works:

- Quartz vibrates at a fixed frequency when electrical current passes through it.
- These vibrations are translated into clock pulses for the motherboard.

